

☒ 1. Computer Forensics Involves

Computer forensics involves a series of **scientific methods and legal practices** to gather and examine digital evidence that is admissible in a court of law. It focuses on:

- Discovering digital evidence
- Investigating and analyzing evidence
- Recovering deleted data
- Maintaining integrity of the data
- Preparing for legal presentation

Tools: EnCase, FTK, Autopsy, Sleuth Kit, Volatility.

☒ 2. Preservation

Preservation is the **second stage** in the forensic process after identification. It involves:

- Creating **exact copies** (bit-by-bit images) of original data
- Using **write blockers** to avoid modifying the original data
- **Hashing** data to verify integrity (e.g., MD5, SHA-256)
- Ensuring **evidence is not tampered** with during investigation

☒ 3. Identification

The process of identifying:

- **Potential evidence sources** (e.g., computers, mobile phones, servers)
- **Relevant data types:** logs, documents, emails, chat logs, deleted files
- The scope of the incident and devices involved

Requires understanding of **file systems, network paths, and user access.**

☒ 4. Extraction

Extraction refers to:

- **Recovering deleted, hidden, or encrypted files**
- **Accessing temporary files, registry keys, and memory dumps**

- Utilizing forensic tools to gather raw data
- Extracting information without altering metadata or timestamps

Common extraction types:

- Logical extraction (file-level)
- Physical extraction (entire device)
- Live extraction (while the system is running)

☒ 5. Documentation

Documentation is essential for:

- Creating a **step-by-step record** of what was done, when, and by whom
- Tracking **tools used**, evidence collected, hash values, and observations
- Ensuring **legal admissibility** through consistent logs
- Supporting reports and possible courtroom testimony

Must be **clear, detailed, and legally sound**.

📄 6. Interpretation

Involves **analyzing extracted data** to determine:

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Timeline of events (what happened and when)

- User actions and intentions
- Data flow and compromise patterns
- Connections between devices, IP addresses, and accounts

Results must be **logically structured and understandable** to non-technical stakeholders.

☒ 7. Goals of Forensics Analysis

The primary goals are:

- Recovering and analyzing evidence
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Maintaining evidence integrity

- Attributing actions to individuals
- Assisting legal proceedings
- Providing intelligence for future prevention

Secondary goals:

- Incident response
 - Internal investigations
 - Audits and policy violations
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❏ 8. Types of Cyber Forensics Techniques

Technique	Description
Disk Forensics	Analyzing hard drives and storage media
Network Forensics	Monitoring and analyzing network traffic
Memory Forensics	Analyzing volatile data in RAM
Mobile Forensics	Investigating smartphones and tablets
Cloud Forensics	Accessing data stored in cloud platforms
Database Forensics	Examining data stored in DBMS systems
Malware Forensics	Reverse-engineering and analyzing malware

❏ 9. Cyber Forensics Procedures

Standard procedures include:

1.
Preparation
2.
Detection of incident
3.
Isolation of affected systems
4.
Preservation of data
5.
Collection and extraction
6.
Analysis and interpretation
7.
Reporting and documentation
8.
Legal proceedings support

Adhering to **standard operating procedures (SOPs)** ensures reliability and admissibility.

☒ **10. Preparation**

Preparation involves:

- **Setting up a forensic lab** with required tools
- **Defining policies and procedures**
- **Ensuring legal authorization**
- **Training staff in incident response**
- **Preparing tools, checklists, and storage**

Preparedness improves speed and quality of investigation.

☒ 11. What to Do Before the Incident

Pre-incident planning includes:

- Establishing a cybersecurity framework
- Defining roles and responsibilities
- Implementing logging and monitoring systems
- Performing risk assessments
- Conducting regular training and awareness programs

❏ 12. Incident Response Plan

A documented plan detailing:

- How to **detect, respond to, and recover** from incidents
 - Key contacts and escalation paths
 - Steps to isolate and preserve evidence
 - Guidelines for communication and containment
 - Links with legal teams and law enforcement
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☒ 13. Incident Response Team

A group of trained personnel responsible for handling security incidents:

Roles:

- **Team Leader** – Coordinates response and resources
 - **Forensic Analyst** – Acquires and analyzes data
 - **System Admin** – Implements containment and recovery
 - **Legal Advisor** – Ensures legal compliance
 - **Communication Officer** – Handles internal/external updates
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☒ 14. Detecting Incidents

Methods include:

- Intrusion detection systems (IDS)
- SIEM tools (e.g., Splunk, AlienVault)
- Unusual behavior analysis
- System logs and alerts
- User-reported issues

Indicators: unauthorized access, system crashes, unusual traffic, changes in file integrity.

☒ 15. Chain of Custody

A **chronological documentation** showing:

- Who handled the evidence
- When and where it was handled
- Under what circumstances

It is **critical for evidence admissibility** in court. Includes:

- Unique evidence IDs
- Dates, times, and signatures
- Location logs and access controls

